

Response to RFI on the Development of an Artificial Intelligence (AI) Action Plan

Submitted by: University of Oklahoma

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Introduction

The University of Oklahoma welcomes the opportunity to provide input on the development of a national Artificial Intelligence (AI) Action Plan. We recognize AI's transformative potential across industries and academia while also emphasizing the need for responsible development, governance, and education. This response addresses key areas of concern and opportunity, particularly within AI research, education, policy, and public perception.

Key Areas Include:

1. AI Research Funding and Infrastructure

A robust research ecosystem is essential for continued AI innovation. Universities serve as critical hubs for fundamental AI research, fostering interdisciplinary collaboration and developing ethical frameworks for AI deployment.

- **Need for Sustainable AI Research Funding:** Federal support for AI research should prioritize long-term funding for universities, including grants that support interdisciplinary AI studies across STEM, social sciences, and humanities.
- **Access to AI Compute Resources:** Universities require access to high-performance computing infrastructure, particularly for training and evaluating AI models. National investments in shared research computing centers could help universities compete with industry in AI development.

2. Bias and Explainability in AI Models

One of the fundamental challenges in AI development is mitigating bias in large language models (LLMs) and other AI systems.

- **Human Expertise in Research Pipelines:** It is essential to integrate domain experts throughout the research process to ensure context-aware model development and to address known issues such as AI hallucinations and misinformation. Human oversight is necessary to validate AI-generated content and prevent factual errors from propagating within academic and professional settings. Developing robust human-computer visual interaction with the models with incremental training is key.
- **AI Model Transparency and Assurance:** AI-generated outputs must be explainable and interpretable. Research should prioritize developing tools and methodologies that allow researchers and practitioners to assess AI decision-making and mitigate risks associated with opaque algorithms. Explorable visualization of the decision making process is key for transparency and trustability.
- **Process-Oriented Approach in Research and Education:** AI efficiency should not be defined solely by labor reduction. Instead, AI should enhance the quality of research and analysis, augmenting human efforts rather than replacing them. This perspective is crucial in ensuring that the adoption of AI supports rather than diminishes the role of graduate students, postdocs, and faculty researchers.

3. AI's Role in Research and Scholarly Integrity

As AI becomes increasingly integrated into academic research, there is concern about the potential for AI-generated content to disrupt scholarly integrity.

- **Preventing AI-Generated Research Cycles:** The increasing prevalence of AI-generated articles, illustrations, and even peer reviews raises concerns about a “self-referential” research pipeline where AI tools generate, review, and summarize their own content. This cycle risks diminishing the reliability and originality of academic discourse. Strict guidelines should be implemented to ensure transparency in AI-assisted research and prevent reliance on unverified AI-generated sources.
- **GenAI in Humanities and Linguistic Research:** AI has the potential to significantly enhance research methods, such as improving handwritten text recognition (HTR) and generating synthetic data for low-resourced languages. These applications should be developed with robust validation mechanisms to ensure accuracy and historical fidelity.

4. AI in Education: Opportunities and Challenges

AI's integration into education presents opportunities but also challenges related to cognitive engagement, academic integrity, and data privacy.

- **Avoiding Cognitive Offloading:** AI should be used as an educational tool that enhances critical thinking rather than replacing intellectual effort. Classroom AI use should not reduce students' engagement to merely generating prompts while AI completes assignments.
- **Emphasizing Learning Process Over Outcomes:** Students often prioritize final grades and completed assignments over learning. AI-enabled education should reinforce the importance of the research and learning process rather than just producing efficient results.
- **Ethical Considerations in AI-Assisted Teaching:** The use of AI in grading and assessments must be carefully considered to avoid automated evaluation loops where AI-generated assignments are graded by AI. The educational system must maintain meaningful human oversight to ensure learning outcomes align with pedagogical goals.
- **Protecting Student Data and Privacy:** With increasing AI adoption, safeguarding student data from unauthorized access and misuse is critical. AI policies should enforce strict data privacy measures in educational applications.

5. AI Workforce Development and Higher Education

Building a well-prepared AI workforce is crucial for the nation's economic and technological growth. Universities play a central role in developing AI curricula, fostering interdisciplinary learning, and equipping students with technical and ethical AI competencies.

- **AI Curriculum and Interdisciplinary Education:** Universities should be key partners in developing AI curricula that prepare students for AI-related careers. This includes ethics, policy, and technical training to create a well-rounded AI workforce.

- **Bridging the Academia-Industry Gap:** Stronger collaboration between universities and industry can help translate academic AI research into real-world applications while maintaining ethical and societal considerations.

6. AI for Public Good and Societal Challenges

AI has the potential to address major societal challenges, from disaster response to healthcare and environmental sustainability. Universities can serve as innovation centers for AI applications that promote public welfare.

- **AI in Public Sector and Civic Applications:** Universities play a role in developing AI applications for public benefit, including AI-driven solutions for disaster response, healthcare equity, environmental monitoring, and more.
- **AI in Public Policy and Governance:** Academic institutions can contribute to AI policy discussions by providing independent, evidence-based research to guide ethical AI regulations.

7. AI Stigmatization and Public Perception

Public perception of AI plays a significant role in shaping regulatory landscapes and adoption. Historical examples, such as nuclear energy and genetically modified organisms (GMOs), demonstrate how technological stigma can hinder progress.

- **Addressing Public Concerns Without Hindering Innovation:** AI has been the subject of polarized debates, with some advocating for extreme restrictions and others warning of catastrophic societal impacts. A balanced, evidence-based approach is essential to mitigate undue fears while addressing legitimate concerns.
- **Managing Risk Perception Through Transparent Communication:** Sensationalist media coverage and regulatory uncertainty can amplify public apprehensions. Government agencies should implement proactive communication strategies to convey AI's benefits while acknowledging and addressing its risks.
- **Ensuring AI's Role in Strengthening the U.S.'s Global Leadership:** Overly restrictive AI policies could drive innovation offshore, potentially leading to weaker safeguards in other countries. The U.S. should aim to foster AI development under a responsible innovation framework that maintains global competitiveness.

Conclusion

The University of Oklahoma emphasizes the need for a balanced AI Action Plan that supports innovation while addressing ethical, educational, and societal challenges. Our recommendations focus on ensuring human expertise remains central in AI research building upon robust, collaborative visual interaction with the AI algorithm training and use, preventing AI-generated research cycles, fostering responsible AI adoption in education, and countering AI stigmatization through transparent public engagement. By implementing thoughtful policies, the U.S. can lead the way in AI development while upholding ethical and scholarly integrity.

We appreciate the opportunity to contribute to this critical national initiative and look forward to further collaboration on responsible AI policy development.